

GLYCOL – The Facts and Cautions - ADEY US

Category: Propylene Glycol Aquatic Chemistry & River Spill Dangers |
Evidence ID: #017

Archived: 2026-10-04 20:13

- **Original URL:** <https://adey.com/us/glycol-the-facts-and-cautions>
 - **Research Category:** Propylene Glycol Aquatic Chemistry & River Spill Dangers
 - **Archival File:** 017_adey_com_us_glycol-the-facts-and-cautions.pdf
 - **Extraction Status:** Full Web Transcript PDF
-

Source Content

Glycol degrades over time, causing the water to become acidic. Acidic water can corrode any metal in a system. In this article, ADEY's Antonia Aldridge covers FAQs, the facts and the cautions when using glycol. Types of Glycol – which one should I use? Firstly, the type of glycol being used in a hydronic system is crucial. Always use propylene. Ethylene is extremely toxic – it's poisonous and human contact with it should be restricted where possible. Propylene has low toxicity levels meaning it's a much safer option. How much Glycol should I add to a system? Boiler manufacturers often advise that glycol should make up no more than 50% of the system's volume, with some HE boiler applications being reduced to 35%. Can system efficiency be affected if too much glycol is added? Yes, glycol has a higher viscosity than water – too much of it can reduce the effectiveness of water flowing through pipework, and too much glycol can also reduce heat transfer capabilities. Both of these aspects together can have a serious impact on heating components such as pumps and heat exchangers, causing them to work harder to maintain system temperature and flow, so much so that it can lead to over-heating and eventually, failure. This is why many boiler manufacturers have warning limitations on the amount of glycol used, because too much can damage components and even result in boiler failures. Before adding any propylene glycol to a system, understand the design life of it. Often, the less expensive propylene glycol holds a shorter design life of 3 – 5 years, whereas the premium propylene glycol holds a design life of roughly up to 15 years. As propylene glycol begins to degrade, it becomes acidic, which has a major impact on key components within the system – attacking metal surfaces such as iron pipework, pumps and heat exchangers, which is why the design life of glycol needs to be seriously considered. A premium glycol contains a higher quality inhibitor which acts as a buffer to keep the system's water pH neutral, but the system should be checked on a regular basis anyway – the ideal level is 7 – 8.5. Once system pH goes below 6.5, inhibitor with buffering agents will need to be added to elevate the pH levels. Can I use a cleaner if Glycol is already in the system? Yes, but test the water before you do. A pH test is the perfect way to find out whether the glycol existing in the system is already degrading. We advise to add ADEY's MC1+ protector as well as cleaner, because this has pH buffering agents that will neutralize pH levels. I've already added/or plan to add pre-mix glycol to the system. Do I need a cleaner and magnetic filter too? Yes. Pre-mix glycol is not designed to lift and capture magnetite – it's designed to prevent system water from freezing during cold winter months, and the inhibitors in pre-mix glycol are used as a buffer to prevent low and high acidity levels. Magnetite, also known as black iron oxide sludge is formed as a result of air oxidizing inside an untreated system. Even in a new system, air is likely to enter and unfortunately, a pre-mix glycol does not mean that a system has been sufficiently treated. A cleaner

contains surfactants and dispersants that surround the magnetic particles, bonding them until they become dislodged from surfaces such as metal pipework, ready to be flushed out of the system. A magnetic dirt filter provides long-term protection. Just because the system has been cleaned and flushed once, doesn't mean that magnetite can still occur. The filter continuously collects magnetic dirt particles, preventing them from circulating through vital narrow waterways and blocking valuable system components such as pumps and heat exchangers. Ideally, a system should be cleaned and flushed before glycol is added, and a magnetic filter should be installed for long-term protection. If the glycol used isn't pre-mix, then inhibitor should be used also.